

THE NO-THREE-IN-A-LINE PROBLEM ON $3 \times n$ GRIDS

*Explicit Construction, and the Enumeration
of Extremal Configurations*

A Research Paper in Combinatorial Geometry

Shiyi Yang

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ABSTRACT

The “no-three-in-line” problem is a classic math puzzle from 1900 that asks how many points you can place on a grid without any three of them forming a straight line. While we know the maximum number of points can never be more than twice the grid’s width, mathematicians still haven’t found a formula to calculate the exact answer for grids of all sizes. This paper tackles a simpler, restricted version of the mystery. Instead of looking at a grid that grows larger in both directions, this study locks the grid’s height at exactly three rows and only lets the number of columns grow. By narrowing the focus this way, the paper takes an unsolved math mystery and completely solves it. We prove that $f(3, n) = \min(2n, 6)$ for every $n \geq 1$, give a single explicit six-point configuration that achieves this bound for every $n \geq 3$ simultaneously, and handle the two small cases $n = 1, 2, 3, 4$ separately. The central tool is a reduction lemma that converts the geometric condition into a purely arithmetic condition on sum-sets, so that checking collinearity on a three-row grid comes down to a single set-disjointness check. Using this reduction, we computationally enumerate every extremal configuration of $\Gamma_{3,n}$ for $3 \leq n \leq 20$, both as raw configurations and after accounting for the grid’s symmetry group.

Keywords: no-three-in-line problem, lattice points, collinearity, arithmetic progressions, sumsets, combinatorial enumeration, symmetry reduction

1 Introduction

Consider an $n \times n$ array of lattice points, n rows by n columns. How many of these points can be selected so that no three selected points are collinear, in any direction: horizontal, vertical, or diagonal? This is the **no-three-in-line problem**. First posed by Henry Dudeney in 1900 and published in his 1917 puzzle collection *Amusements in Mathematics* [1], the problem remains formally unsolved [2] though it is easy enough to state that anyone can understand the question.

Let Γ_n denote the $n \times n$ grid of integer points (x, y) satisfying $0 \leq x, y \leq n - 1$ and $f(n)$ be the maximum cardinality of a valid configuration containing no three collinear points. A fundamental application of the Pigeonhole Principle establishes that $f(n) \leq 2n$. Because each of the n rows forms a horizontal line, no row can contain more than two points without violating the collinearity constraint. Summing this maximum capacity across all n rows yields the upper bound $f(n) \leq 2n$ [3]. The central open question of the field is whether this bound is exactly achievable for all values of n . Exhaustive computer searches have confirmed that the bound is attained for every grid size checked to date [4].

The strongest methods for finding the minimum number of points that can be placed on a grid without forming lines rely on highly advanced algebra, such as complex number theory and finite fields [5], [6]. Rather than diving into these deep algebraic concepts, this paper simplifies the problem to make it completely manageable. By locking the grid's height at exactly three rows and letting the number of columns grow infinitely, the paper avoids complicated algebraic theory and provides a straightforward, complete solution.

Fixing the row dimension alters the structural dynamics of the problem. In classical $n \times n$ square grid, the row and column capacity bounds coincide at $2n$, leaving it open whether this shared bound is asymptotically achievable. Conversely, on a $3 \times n$ grid ($\Gamma_{\{3,n\}}$), the row capacity locks the trivial upper bound at 6 for all $n \geq 3$, whereas the column capacity $2n$ grows indefinitely. Thus, the bottleneck for almost all n is entirely dictated by the row capacity.

This restriction turns an open problem into a closed one without making it trivial. This paper's primary contributions include an algebraic reduction and an exact enumeration of every extremal configuration $\Gamma_{3,n}$ up to $n = 20$.

This paper is organized as follows. Section 2 defines few important terms and the function $f(m, n)$. Section 3 establishes the trivial upper bound $f(3, n) \leq \min(2n, 6)$ by counting row and column capacities with Pigeonhole Principle. Section 4 translates the geometric condition “no three points are collinear” into three purely arithmetic conditions on the column-sets A, B, C of the three rows, culminating in the Full Elimination criterion (Theorem 4.4). Section 5 proves the main theorem, $f(3, n) = \min(2n, 6)$ for every $n \geq 1$, by treating the small cases $n = 1, 2$ directly and then exhibiting a single universal six-point construction that works for every $n \geq$

3 simultaneously. Section 6 develops the group theory needed to compare configurations that are mirror images or rotations of one another and determines the full symmetry group of the $3 \times n$ grid (Lemma 6.2). Finally, Section 7 combines the arithmetic criterion of Section 4 with the symmetry group of Section 6 to computationally enumerate every extremal configuration for $3 \leq n \leq 20$, reporting both the raw count $R(n)$ and the symmetry-reduced count $N(n)$, and explains in detail how duplicate configurations are accounted for.

2 Preliminaries

2.1 Definitions and Notation

We begin by establishing the formal definitions and geometric notations.

Definition 2.1 (Rectangular Grid). For positive integers m, n , the $m \times n$ *grid* is the set of integer points

$$\Gamma_{m,n} = \{(x, y) \in \mathbb{Z}^2 : 0 \leq x \leq n-1 \wedge 0 \leq y \leq m-1\}.$$

We call y the *row index* (there are m rows, $y = 0, \dots, m-1$) and x the *column index* (there are n columns, $x = 0, \dots, n-1$). When $m = n$ we write Γ_n , recovering the classical square grid.

Definition 2.2 (Collinearity and General Position). Let $P_1 = (x_1, y_1)$, $P_2 = (x_2, y_2)$, $P_3 = (x_3, y_3)$ be three pairwise distinct points, i.e. $P_1 \neq P_2$, $P_2 \neq P_3$, and $P_1 \neq P_3$. These three points are *collinear* if they lie on a common line, which holds if and only if

$$(x_2 - x_1)(y_3 - y_1) = (x_3 - x_1)(y_2 - y_1).$$

If two of the three points coincide, this equation holds trivially (both sides vanish), but such is not considered a collinear triple of three distinct points, and is excluded from consideration throughout this paper. A set $S \subseteq \mathbb{R}^2$ is in *general position* if no three pairwise distinct points of S are collinear.

Definition 2.3 (The No-Three-in-a-Line Function). For positive integers m, n , define

$$f(m, n) = \max\{|S| : S \subseteq \Gamma_{m,n}, S \text{ is in general position}\}.$$

When $m = n$ this recovers the classical function $f(n) = f(n, n)$ studied by Dudeney, Guy, Kelly, and others [1], [3]. This paper determines $f(3, n)$ exactly for every n .

Since $\Gamma_{m,n}$ is finite, the maximum in Definition 2.3 is taken over a finite, nonempty collection of subsets, so $f(m, n)$ is always well-defined.

2.2 Collinearity via Slopes

Alternatively, collinearity can be framed through the pairwise slopes. Two points $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ with $x_1 \neq x_2$ determine a slope:

$$\text{slope}(P_1, P_2) = \frac{y_2 - y_1}{x_2 - x_1}$$

If $x_1 = x_2$, then the line segment is vertical.

For three points with pairwise distinct x coordinates, they are collinear if and only if they lie on a single straight line, which is displayed by the following formula:

$$\text{slope}(P_1, P_2) = \text{slope}(P_2, P_3)$$

This equation acts as the basis for the non-vertical part of the reduction carried out in Section 4: once a grid has only three rows, every relevant slope condition can be written down explicitly and compared directly, rather than checked case by case for every possible line.

3 The Trivial Bound for $3 \times n$ Grids

Before analyzing specific point configurations, we can establish a strict upper limit on the maximum number of points a valid configuration can hold: $f(3, n)$. This is called the trivial bound since it relies solely on the dimensions of the grid rather than any complex geometric properties.

Theorem 3.1 *For every $3 \times n$ grid where $n \geq 1$, the maximum number of points satisfies:*
 $f(3, n) \leq \min(2n, 6)$.

The same set of points is counted two different ways, once by column and once by row. By using the pigeonhole principle each time, the smaller of the two counts is taken, since both bounds must hold at once.

Proof. Let $S \subseteq \Gamma_{3,n}$ be in general position.

(1) *Bound by columns.* Suppose a configuration contains more than $2n$ points. The grid has exactly n columns, so at least one column must contain three or more points. Since any three points in a single vertical column are inherently collinear, the configuration becomes invalid. Therefore, the total number of points cannot exceed $2n$.

(2) *Bound by rows.* Similarly, consider the rows. The grid contains exactly 3 horizontal rows. If any single row contains three or more points, those points are collinear. Consequently, each row can hold a maximum of 2 points. Multiplying the 3 available rows by this 2-point limit yields a maximum capacity of 6 points, regardless of how large n becomes.

Both bounds hold simultaneously, so $|S| \leq \min(2n, 6)$. $\square$

Notably, Theorem 3.1 does not account for constraints imposed by diagonal or arbitrary rational slopes. We address these remaining geometric constraints in following Section 4 by reducing them to a tractable arithmetic condition.

4 Reduction to an Arithmetic Condition

To systematically analyze valid configurations on a $3 \times n$ grid, we can reduce the geometric constraint, that no three selected points are collinear, into a purely algebraic framework.

Let $S \subseteq \Gamma_{3,n}$ be a given subset of lattice points. Because the grid has exactly three rows, every point $P = (x, y) \in S$ must satisfy $y \in \{0, 1, 2\}$.

Let the column indices used in each respective row be:

$$\begin{aligned} A &= \{x : (x, 0) \in S\} \subseteq \{0, \dots, n-1\}, \\ B &= \{x : (x, 1) \in S\} \subseteq \{0, \dots, n-1\}, \\ C &= \{x : (x, 2) \in S\} \subseteq \{0, \dots, n-1\}, \end{aligned}$$

Under this representation, any collinear triple of distinct points in S must fall into one of three categories based on how many rows the triple intersects:

1. All three points lie in a single horizontal row.
2. All three points lie in a single vertical column.
3. The three points span across different rows and columns along a diagonal line.

The following lemmas establish the necessary and sufficient algebraic conditions to eliminate each type of collinearity stated above.

4.1 Horizontal Elimination

Lemma 4.1. *A configuration S contains no horizontal collinear triples if and only if*

$$|A| \leq 2, |B| \leq 2, |C| \leq 2$$

Proof. By definition, a horizontal line in $\Gamma_{\{3,n\}}$ is uniquely determined by a fixed row index $y \in \{0, 1, 2\}$. If any row contains three or more points, those points share an identical y -coordinate, meaning they are mutually collinear. Conversely, if the cardinality of each column-set A , B , and C is at most 2, no single row possesses a third point to form a horizontal triple. $\square$

4.2 Vertical Elimination

Lemma 4.2. *A configuration S contains no vertical collinear triples if and only if*

$$A \cap B \cap C = \emptyset$$

Proof. A vertical line in $\Gamma_{3,n}$ is uniquely determined by its column index $x \in \{0, 1, \dots, n-1\}$. Since $\Gamma_{3,n}$ has exactly three rows, a vertical collinear triple consists precisely of the three points in a single column, namely one point from each of rows 0, 1, and 2.

Such a triple exists if and only if there is some column index x satisfying $x \in A$, $x \in B$, and $x \in C$, equivalently, $x \in A \cap B \cap C$. Hence, the configuration contains no vertical collinear triples if and only if $A \cap B \cap C = \emptyset$. $\square$

4.3 Slanted Elimination

Lemma 4.3. *A configuration S contains no slanted collinear triples if and only if*

$$(A + C) \cap 2B = \emptyset,$$

where $A + C$ is the sumset of A and C , and $2B = \{2b : b \in B\}$.

Proof. Suppose three distinct points $P_0, P_1, P_2 \in S$ form a slanted collinear line. By Lemmas 4.1 and 4.2 (stated earlier), they cannot share a row or a column; thus, they must possess strictly distinct row coordinates and strictly distinct column coordinates. Because there are exactly three rows available, a slanted collinear triple must feature exactly one point in each row. Without loss of generality, let these points be ordered by their row coordinates:

$$P_0 = (a, 0) \in \text{Row } 0 \Rightarrow a \in A$$

$$P_1 = (b, 1) \in \text{Row } 1 \Rightarrow b \in B$$

$$P_2 = (c, 2) \in \text{Row } 2 \Rightarrow c \in C$$

Applying the standard cross-multiplication criterion for collinearity from Definition 2.2 gives

$$(x_1 - x_0)(y_2 - y_0) = (x_2 - x_0)(y_1 - y_0).$$

Substituting the coordinates yields

$$(b - a)(2 - 0) = (c - a)(1 - 0).$$

After simplifying, it yields:

$$2b = a + c.$$

Thus, three points occupying the three rows are collinear if and only if their column indices form a three-term arithmetic progression, with the middle term belonging to row 1 (B) and the endpoints belonging to rows 0 (A) and 2 (C).

Therefore, a slanted collinear triple exists if and only if

$$2B \cap (A + C) \neq \emptyset.$$

Equivalently, the configuration contains no slanted collinear triples if and only if

$$(A + C) \cap 2B = \emptyset. \square$$

4.4 Full Elimination

Theorem 4.4. *A subset $S \subseteq \Gamma_{3,n}$ is a valid no-three-in-line configuration if and only if the following three independent structural conditions are satisfied:*

$$\max(|A|, |B|, |C|) \leq 2,$$

$$A \cap B \cap C = \emptyset,$$

$$(A + C) \cap 2B = \emptyset.$$

Proof. Every potential collinear triple in $\Gamma_{3,n}$ is of exactly one of the following three types: horizontal, vertical, or slanted (crossing all three rows).

By Lemmas 4.1, 4.2, and 4.3, these three cases are characterized respectively by the three

conditions. Since these three classes of collinear triples are mutually exclusive and collectively exhaustive, the configuration contains no three collinear points if and only if all three conditions hold simultaneously. $\square$

4.5 Remark of the Structural Limitations for Higher Dimensions

The reduction in Theorem 4.4 is unique to the $3 \times n$ grid. When the row dimension is restricted to $m = 3$, any slanted collinear triple is structurally forced to intersect every available row exactly once. This ensures that the row indices always form a fixed, symmetric spacing ($y = 0, 1, 2$), which directly yields the arithmetic progression $2b = a + c$.

For grids with $m \geq 4$, this system breaks down. A slanted line can span three points across four or more rows without intersecting every row, or it can feature non-uniform row steps. Consequently, eliminating slanted lines on larger grids cannot be compressed into a single sum-set condition. This explains why the $3 \times n$ case serves as a uniquely tractable sandbox.

5 Main Theorem and Explicit Construction

This section will present the primary findings of the paper: the exact maximum number of points we can place on a 3-row grid without any three of them lining up.

Theorem 5.1 (The Main Rule). *For any grid with 3 rows and n columns, the maximum number of points we can choose without creating a straight line of three points, written as $f(3, n)$, is exactly equal to whichever is smaller: $2n$ or 6.*

Specifically:

- (1) For 1 column ($n = 1$), we can place at most 2 points.
- (2) For 2 columns ($n = 2$), we can place at most 4 points.
- (3) For 3 or more columns ($n \geq 3$), we can always place exactly 6 points, no matter how wide the grid finally gets.

Earlier in the paper, Theorem 3.1 proved that we can never have more than $2n$ or 6 points because of basic grid capacities (for instance, a 3-row grid can only hold a maximum of 2 points per row, meaning $3 \times 2 = 6$ points max). However, proving a strict upper limit doesn't automatically mean that limit is actually possible to reach in reality. To prove the rule is true, we have to show real, working examples for every size of n .

Before solving the problem for massive grids, we must handle the smallest cases first.

5.1 Small Cases: Grids with $n = 1$ and $n = 2$ Columns

For $n = 1$, the grid $\Gamma_{3,1}$ is a single column of 3 points: $(0,0), (0,1), (0,2)$. These points are trivially collinear with a shared vertical column.

Selecting all 3 points will break the rule. However, if we only pick 2 points, it is physically impossible to break the rule since we need at least 3 points to make a forbidden triple. Therefore, the most points we can successfully place is 2. This matches the result we derived from our formula.



Figure 1: Case for $n = 1$

For $n = 2$, the grid has 3 rows and 2 columns, giving us 6 possible spots. Our formula states the limit should be $\min(2(2), 6) = 4$ points. We can reach this maximum by placing as follows, giving us 4 coordinates: $(0,1), (1,1), (0,2)$ and $(1,2)$.

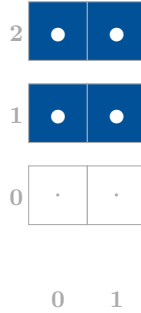


Figure 2: Case for $n = 2$

5.2 Blueprint for ALL Large Grids ($n \geq 3$)

Once a grid has 3 or more columns, the formula locks the maximum capacity at exactly 6 points. Remarkably, we do not need to invent a new pattern for a 4-column grid, a 5-column grid, or a 100-column grid. A single, universal 6-point blueprint works for every single one of

them. It achieves this by packing all 6 dots into the very first three columns (Columns 0, 1, and 2) and leaving the rest of the grid completely empty.

Proposition 5.2 (Universal Construction). *For any grid where the number of columns $n \geq 3$, we select the following column positions for each of our three rows:*

- *Row 0 (Bottom): Choose Columns 1 and 2 - Points: $(1,0)$ and $(2,0)$*
- *Row 1 (Middle): Choose Columns 0 and 2 - Points: $(0,1)$ and $(2,1)$*
- *Row 2 (Top): Choose Columns 0 and 1 - Points $(0,2)$ and $(1,2)$*

This gives us a total of 6 points. No matter how wide the grid grows, these 6 points will never form a straight line of three.

Proof.

1. By Definition 2.1, any point configuration S within the rectangular lattice framework $\Gamma_{\{3,n\}}$ must restrict its column indices to the domain $\{0, 1, \dots, n-1\}$. Inspecting the coordinate sets mapping to each row, we observe that:

$$A, B, C \subseteq \{0, 1, 2\}$$

Given the hypothesis $n \geq 3$, it follows by nesting that $\{0, 1, 2\} \subseteq \{0, 1, \dots, n-1\}$. Consequently, all assigned coordinates lie strictly within the physical boundaries of the lattice $\Gamma_{\{3,n\}}$, ensuring that the configuration is well-defined.

2. Because the horizontal rows $y = 0$, $y = 1$, and $y = 2$ form mutually disjoint geometric subsets of $\Gamma_{\{3,n\}}$, the total cardinality of the configuration is exactly the sum of the cardinalities of the individual row-sets:

$$|S| = |A| + |B| + |C| = 2 + 2 + 2 = 6$$

This explicitly attains the structural capacity upper bound $\min(2n, 6) = 6$ proved above.

3. In order to verify that no three selected points are collinear, we evaluate the configuration against the three independent, mutually exclusive elimination criteria defined in Theorem 4.4:

- **Horizontal Elimination (Lemma 4.1):** A configuration is free of horizontal collinear triples if and only if no individual row exceeds a capacity of 2 points. Since $|A| = 2$, $|B| = 2$, and $|C| = 2$, the constraint $\max(|A|, |B|, |C|) \leq 2$ holds identically.

- **Vertical Elimination (Lemma 4.2):** Vertical collinearity requires three points to share a common column across all three rows. This condition is negated if and only if the mutual intersection of the row sets is empty. Computing this intersection yields:

$$A \cap B \cap C = \{1, 2\} \cap \{0, 2\} \cap \{0, 1\} = \emptyset$$

As the intersection is empty, no vertical collinear triples can exist.

- **Slanted Elimination (Lemma 4.3):** By Lemma 4.3, the configuration contains no slanted collinear triples exactly when the sumset of the outer rows ($A + C$) shares no element with the doubled middle row ($2B$). We check this directly.

We compute the sumset $A + C$ by adding every element of A to every element of C :

$$A + C = \{a + c : a \in A, c \in C\} = \{1 + 0, 1 + 1, 2 + 0, 2 + 1\} = \{1, 2, 3\}$$

We then double each element of the middle row set B :

$$2B = \{2b : b \in B\} = \{2 \times 0, 2 \times 2\} = \{0, 4\}$$

Evaluating the set-theoretic intersection of the resulting sets reveals:

$$(A + C) \cap 2B = \{1, 2, 3\} \cap \{0, 4\} = \emptyset$$

Because $(A + C) \cap 2B = \emptyset$, it is mathematically impossible for any three points to form a three-term arithmetic progression across the row indices.

As all elimination criteria are satisfied, the configuration S is proven to be in general position. Because neither the sumset $A + C$ nor the doubled set $2B$ depends on the column dimension parameter n , this explicit configuration remains universally valid for all $n \geq 3$.

□

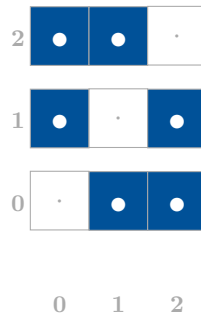
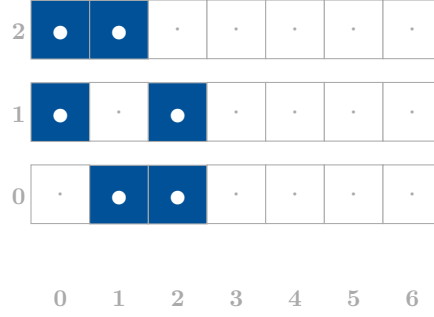


Figure 3: Case for $n = 3$

Figure 4: Case for $n = 7$

The two figures above explicitly display the verification of Theorem 5.1. The case $n = 7$'s construction uses only columns 0, 1, 2, independent of n , matching the upper bound $\min(2n, 6) = 6$ in every case $n \geq 3$.

Remark 5.3. *The universal explicit construction of a fixed 3-row grid has a huge difference with Dudeney's classical square problem Γ_n :*

- *In $n \times n$ square grid, any configuration attempting to reach the upper bound of $2n$ points is bound by tight constraints. By the Pigeonhole Principle, every single row and column must contain exactly 2 points. Leaving even one column empty would cause global capacity collapse.*
- *Conversely, on our $3 \times n$ grid where $n > 3$, the global capacity locks permanently at 6 points due to the rigid row constraints 3×2 . Thus, additional column width yields space abundance without increasing the maximum point.*

6 The Symmetry Group of the $3 \times n$ Grid

6.1 Motivation

In Section 5, we found valid ways to place six points on a $3 \times n$ grid in general position, meaning no three points are collinear. Because geometric transformations like reflections and rotations preserve the straightness of lines, applying them to a valid layout always creates another valid layout. To avoid overcounting these simple flips and rotations as entirely new solutions in Section 7, we must treat them as equivalent orientations of the same underlying pattern. We can achieve accurate count by using a symmetry group to identify and group equivalent configurations.

Fundamental algebraic concepts used throughout this section are introduced below, following the standard treatments in [7] and [8]:

(1) **Set.** A set is a well-defined collection of distinct objects. In this paper, the set $S_{3,n}$ denotes the collection of all maximum-sized (extremal) valid no-three-in-line configurations on a $3 \times n$ grid.

(2) **Group.** A group is a mathematical structure consisting of a set of operations (such as reflections/rotations) equipped with a binary operation, called composition, satisfying the following four axioms [8]:

- **Closure:** The composition of any two group elements is also an element of the group.
- **Associativity:** For any group elements A , B , and C :

$$(A \circ B) \circ C = A \circ (B \circ C)$$

- **Identity:** There exists an identity element, denoted by id , such that composing it with any group element leaves that element unchanged.
- **Inverse:** Every group element has an inverse within the group. For example, reflecting a grid twice across the same axis returns it to its original configuration.

(3) **Group Action.** A group action is a rule describing how each element of a group acts on the elements of a set. Each symmetry operation maps every point in a configuration to a new position, producing another configuration that also belongs to $S_{3,n}$.

(4) **Bijection:** This is a function that matches elements between two sets perfectly. It is both one-to-one (injective), meaning no two different starting points map to the exact same ending point, and onto (surjective), meaning every possible ending point is mapped to by exactly one starting point. In the context of a grid symmetry, a bijection acts as a strict rearrangement or relabeling that never duplicates or omits any grid locations.

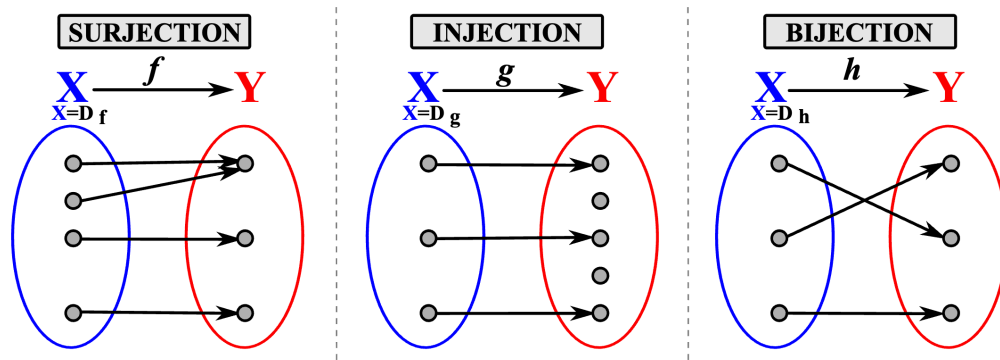


Figure 5: Visualization of injection, surjection, and bijection. Diagram reproduced from the Wikipedia article on bijections [9].

Definition 6.1 (Grid Symmetries). A *symmetry* of the grid $\Gamma_{3,n}$ is a bijection $\Gamma_{3,n} \rightarrow \Gamma_{3,n}$ that preserves the line structures, mapping any three collinear points to three collinear points, and any three non-collinear points to three non-collinear points.

Because a grid symmetry perfectly preserves collinearity and non-collinearity, it can never transform a valid configuration into an invalid one, or vice versa. Therefore, if a configuration is valid before applying the symmetry, it remains valid afterwards.

6.2 The Generating Symmetries H and V

In previous sections, we define the grid $\Gamma_{3,n}$ using integer coordinates (x, y) , where x represents the column index ($0 \leq x \leq n-1$) and y represents the row index ($0 \leq y \leq 2$). When the number of columns n does not equal 3, the grid forms a non-square rectangle. Geometrically, a non-square rectangle possesses exactly four rigid symmetries: the identity operation, a horizontal reflection, a vertical reflection, and a combined horizontal-vertical reflection.

We can write out these primary operations explicitly using precise coordinate transformations:

- **Horizontal Reflection H :** This mapping reverses the horizontal ordering of the columns. Column 0 is mapped to column $n-1$, column 1 is mapped to column $n-2$.

$$(x, y) \mapsto (n-1-x, y) \quad (y \text{ remain unchanged}),$$

- **Vertical Reflection V :** This mapping swaps the outer rows. The bottom row $y=0$ maps to the top row $y=2$, the top row $y=2$ maps to the bottom row $y=0$ with the middle row $y=1$ perfectly fixed since

$$V : (x, y) \mapsto (x, 2-y) \quad (x \text{ remain unchanged}).$$

To better clarify how these operations affect the point sets, we illustrate their action on the universal six-point construction from Proposition 5.2 within a grid of $n=7$ columns.

The original configuration consists of the following coordinate set:

$$S = \{(1, 0), (2, 0), (0, 1), (2, 1), (0, 2), (1, 2)\}$$

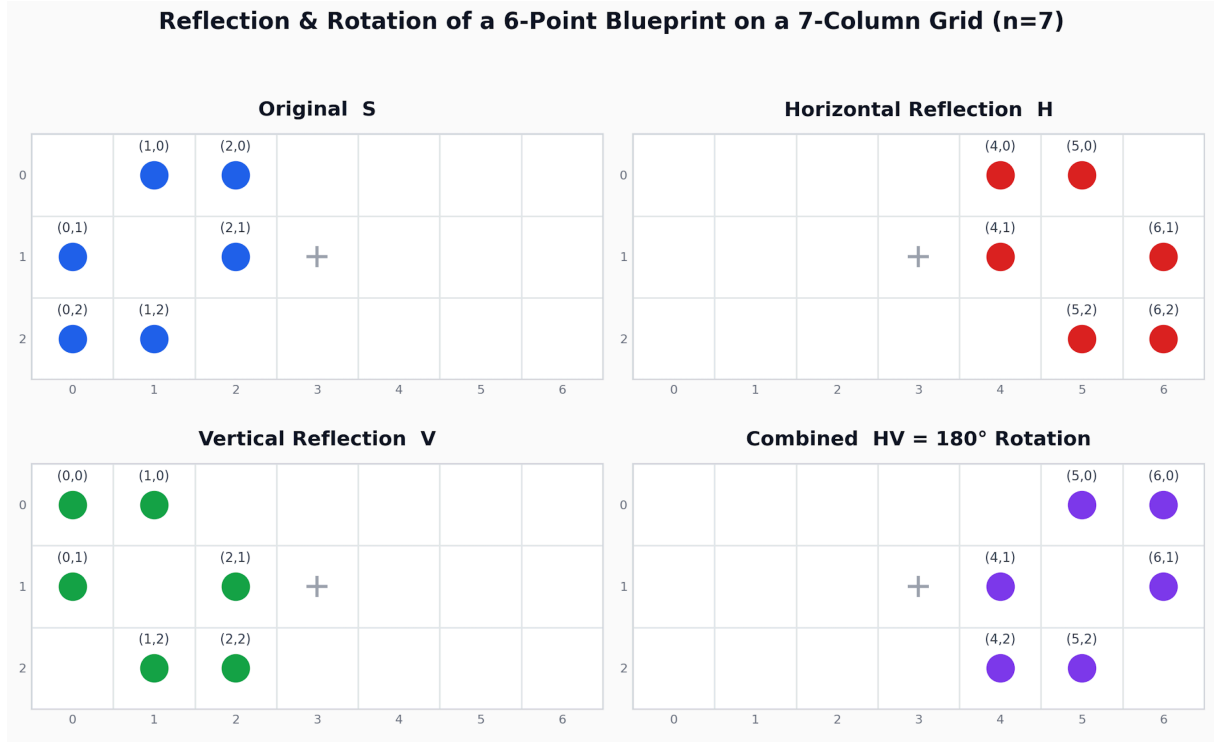


Figure 6: The visualization of grid transformation both vertically and horizontally

- Applying the horizontal reflection H when $n = 7$ replaces each x-coordinate with $7 - 1 - x = 6 - x$. This yields the transformed set $\{(5, 0), (4, 0), (6, 1), (4, 1), (6, 2), (5, 2)\}$, which effectively shifts the entire layout from the left side of the grid to the right side.
- Applying the vertical reflection V replaces each y-coordinate with $2 - y$. This yields the set $\{(1, 2), (2, 2), (0, 1), (2, 1), (0, 0), (1, 0)\}$, which turns the configuration upside down.
- If we perform both operations, we can use function composition to find the combined mapping. Applying V changes (x, y) to $(x, 2 - y)$. Then applying H to this new coordinate pair transforms the horizontal component:

$$HV(x, y) = (n - 1 - x, 2 - y)$$

Algebraically, this combined transformation negates both coordinates relative to the center of the grid, located at $(\frac{n-1}{2}, 1)$. Equivalently, every point is mapped to the point directly opposite the center. In coordinate geometry, any mapping satisfying

$$f(x_c + \Delta x, y_c + \Delta y) = (x_c - \Delta x, y_c - \Delta y)$$

is precisely a 180° rotation about the point (x_c, y_c) . Therefore, the composition HV is the half-turn rotation obtained by composing the horizontal and vertical reflections.

6.3 The Structure of Symmetry Group

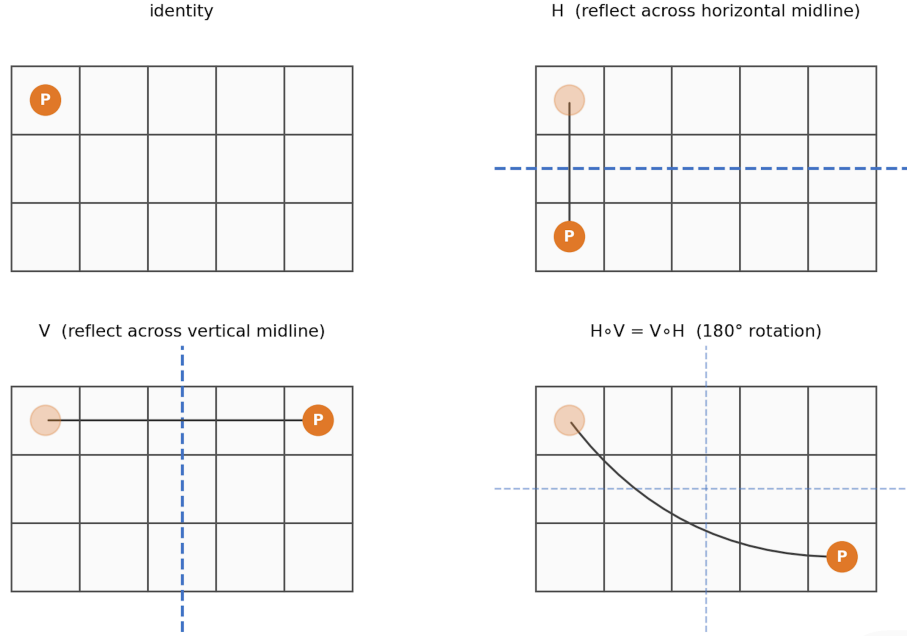
We can summarize the complete set of transformations of the grid using the algebraic framework. Before stating the core lemma, there are few terms that must be defined.

- **Isomorphic:** Two groups are isomorphic if they are structurally identical, even if their individual elements look completely different [7].
- **The Klein Four Group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$:** The group which contains four elements where every single element acts at its own inverse [7]. If we apply any transformation in the group twice in a row, we return exactly to the starting position ($H \circ H = \text{id}$ and $V \circ V = \text{id}$). When any two non-identity transformations are combined, the third one ($H \circ V = HV$) yields.
- **Dihedral Group D_4 :** All possible symmetries of a regular square, including 8 total operations: 4 distinct rotations (0° , 90° , 180° , and 270°) and 4 distinct reflections (one horizontal axis, one vertical axis, and two diagonal axes) [10].

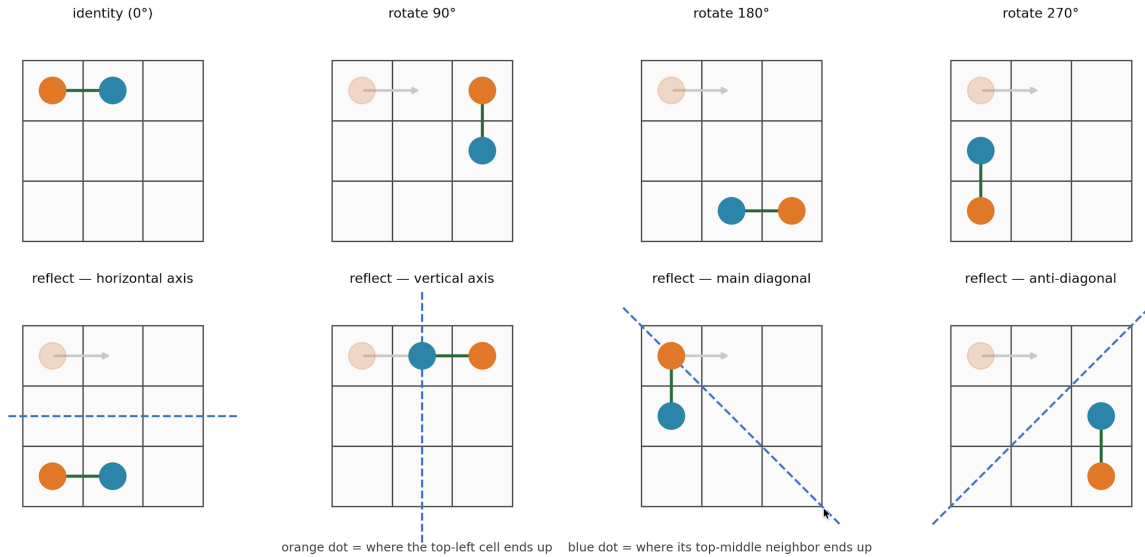
Lemma 6.2. *For all grids where $n \neq 3$, the symmetry group of $\Gamma_{3,n}$ is $G = \{\text{id}, H, V, HV\}$. This group has an algebraic structure isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (the Klein Four-Group). However, when $n = 3$, the grid is a 3×3 square and has the larger symmetry group D_4 (dihedral group), containing 8 total elements.*

The logic behind Lemma 6.2 relies on a basic principle of discrete geometry: any structural symmetry of a grid must map corners to corners [10]. Corners are unique because they are the only points on the grid boundary where the perimeter changes direction sharply by 90 degrees. Because a grid symmetry is required to fully preserve straight lines and distances, it cannot create new corners or destroy existing ones. Thus any valid symmetry is completely fixed by how it chooses to rearrange the four outer corners of the grid.

- **The Rectangular Case ($n \neq 3$):** In a true non-square rectangle, the width and the height are unequal. To preserve rectangle, any corner rearrangement must keep short sides mapping to short sides and long sides mapping to long sides. There are only 4 corner arrangements that satisfy this restriction: doing nothing (id); flipping left-to-right (H); flipping top-to-bottom (V); or rotating by 180° (HV).

Figure 7: Symmetries of $\Gamma_{3,n}$ for $n \neq 3$

- The Square Case ($n = 3$): When the grid is a 3×3 square, rotating the grid by 90° clockwise or counterclockwise keeps the square identical. Because this operation preserves lines, its powers (90° , 180° , 270°) and its combinations with reflections generate the full 8 symmetries of the dihedral group D_4 .

Figure 8: Symmetries of 3×3 square (Dihedral Group)

7 Enumeration of Extremal Configurations

While Theorem 5.1 proves that a valid six-point configuration can always be constructed on any grid where $n \geq 3$, it does not imply that there is only one unique solution. In reality, a single grid can host many duplicates created by rotating or mirroring an existing layout.

Counting every valid coordinate set independently will lead to overcounting of the total number of solutions (up to 4 times on a rectangular grid / up to eight times on a square grid). By establishing the symmetry group G in section 6, we gain the mathematical tool necessary to identify and bundle these duplicate arrangements together. In Section 7, we will discuss exactly how many different ways can we place these six points on a grid of a given size?

7.1 Computational Method

By the Full Elimination (established earlier in Theorem 4.4), we have a clear blueprint for what a valid 6-point layout looks like. Instead of checking lines, angles or slopes, the theorem translates the problem into simple arithmetic rules using three sets of column numbers:

1. Set A: The column numbers used in the bottom row (Row 0).
2. Set B: The column numbers used in the middle row (Row 1).
3. Set C: The column numbers used in the top row (Row 2).

Because we want exactly 6 points total, and no single row can hold more than 2 points, each of these three sets must contain exactly 2 numbers chosen from the available columns (labeled 0 to $n - 1$). Theorem 4.4 tells us that these points will be perfectly safe from diagonal lines as long as the math condition $(A + C) \cap 2B = \emptyset$ is met. Therefore, if we add every number in Set A to every number in Set C, none of those sums can be equal to double any number in Set B. A two-stage Python program (shown below, with the arithmetic test implemented in the function `master_lemma_ok`) can be used to solve the problem with the help of this algebraic shortcut.

```
from itertools import combinations

def collinear(p1, p2, p3):
    # Exact integer collinearity test (Definition 2.2). Using the
    # cross-multiplied form avoids floating-point slope divisions.
    x1, y1 = p1; x2, y2 = p2; x3, y3 = p3
    return (x2 - x1) * (y3 - y1) == (x3 - x1) * (y2 - y1)

def is_general_position(S):
    # Brute-force check: test every one of the  $C(6,3) = 20$  triples
    # of points in S; S is valid only if no triple is collinear.
    return all(not collinear(a, b, c) for a, b, c in combinations(S, 3))

def master_lemma_ok(A, B, C):
```

```

# Slanted Elimination (Lemma 4.3): (A + C) intersect 2B must be empty.
AC = {a + c for a in A for c in C} # sumset A + C (at most 4 values)
twoB = {2 * b for b in B}          # doubled middle row (at most 2 values)
return AC.isdisjoint(twoB)          # True means no slanted triple exists

def points_from_sets(A, B, C):
    # Convert the three column-sets back into the 6 grid points (x, y),
    # with A in row 0 (bottom), B in row 1 (middle), C in row 2 (top).
    return [(a, 0) for a in A] + [(b, 1) for b in B] + [(c, 2) for c in C]

# Sanity check: the algebraic test (Lemma 4.3) must agree with the
# geometric brute-force test on known examples before we trust it.
examples = [
    ({1,2}, {0,2}, {0,1}), # Proposition 5.2 universal construction
    ({0,1}, {2,3}, {0,1}), # another valid 6-point configuration
]
for A, B, C in examples:
    S = points_from_sets(A, B, C)
    assert master_lemma_ok(A, B, C) == is_general_position(S)
print("Master Lemma verified against brute force on examples.")

# Enumeration: raw count R(n) and symmetry-reduced count N(n)
def enumerate_extremal(n):
    cols = list(range(n))
    pairs = list(combinations(cols, 2)) # all C(n,2) two-element column-sets
    raw = 0
    canon_set = set()
    for A in pairs:
        for B in pairs:
            for C in pairs:
                if master_lemma_ok(A, B, C):
                    raw += 1 # Stage 1: count this raw configuration
                    # Stage 2: apply the symmetry group G = {id, H, V, HV}
                    # of Lemma 6.2, then keep the smallest image as the
                    # canonical label (ID tag) of this configuration's family.
                    S = tuple(sorted(points_from_sets(A, B, C)))
                    H = tuple(sorted((n-1-x, y) for x, y in S)) # left-right flip
                    V = tuple(sorted((x, 2-y) for x, y in S)) # top-bottom flip
                    HV = tuple(sorted((n-1-x, 2-y) for x, y in S)) # 180-degree turn
                    canon_set.add(min(S, H, V, HV))
    return raw, len(canon_set) # returns the pair (R(n), N(n))

if __name__ == '__main__':
    for n in range(3, 21):
        R, N = enumerate_extremal(n)
        print(f'n={n:2d} R(n)={R:9d} N(n)={N:8d}')

```

Stage 1: Counting Raw Configurations. First, the program calculates every possible way to choose 2 column numbers for a single row; in the code this is the single line `pairs = list(combinations(cols, 2))`. The number of ways to pick 2 items out of n available options is written as $\binom{n}{2}$. Since we have three separate rows A , B , and C , the computer loops through every single combination for all three rows (the triple for loop over `pairs` in `enumerate_extremal`). The total number of combinations is:

$$\text{Total Triples} = \binom{n}{2} \times \binom{n}{2} \times \binom{n}{2} = \binom{n}{2}^3$$

For a grid with $n = 20$ columns, this is $190^3 \approx 6.9$ million possible combinations, small enough for the computer to check in seconds. For every combination, the computer performs the full elimination lemma's arithmetic test inside the function `master_lemma_ok(A, B, C)`:

(1) It builds the set of all possible sums of an element from A and an element from C , via $AC = \{a + c \text{ for } a \text{ in } A \text{ for } c \text{ in } C\}$.

(2) It doubles the numbers in Set B (producing at most 2 numbers), via $twoB = \{2 * b \text{ for } b \text{ in } B\}$.

(3) It checks if the two sets share any numbers in common, using `AC.isdisjoint(twoB)`. If the sets share absolutely no numbers, the arrangement is valid, and the computer adds it to our Raw Total (the counter `raw` in the code), labeled as $R(n)$.

Stage 2: Filtering Out Symmetric Duplicates. To find the exact number of unique configurations, $N(n)$, we must filter out duplicates resulting from rotation or mirror.

To accomplish this, the code uses a trick called **canonical labeling**:

- For every valid raw layout found in Stage 1, the program generates its coordinate set S of 6 points (the line `S = tuple(sorted(points_from_sets(A, B, C)))`).
- It applies all the possible grid symmetries to that layout to find all its “mirror images” and “rotations”: in the code, the three lines computing H , V , and HV produce exactly the images of S under the three non-identity elements of the symmetry group G from Lemma 6.2.
- Finally, the program compares these variations and selects the mathematically “smallest” coordinate set to serve as the absolute ID tag for that entire family of shapes `canon_set.add(min(S, H, V, HV))`.

Because a configuration and all its flipped or rotated versions share the exact same ID tag, the computer can find $N(n)$ simply by counting the total number of unique tags (`len(canon_set)`). This approach guarantees an exact count of genuinely distinct patterns, completely free of duplication or estimation errors.

It is worth explaining in more detail how this procedure accounts for symmetry. The family of all layouts obtainable from a configuration S by applying elements of G is called the *orbit* of S under the group action [8]. On a rectangular grid, G has 4 elements, so every orbit contains *at most* 4 distinct layouts, but not always exactly 4. If a configuration happens to be symmetric itself, say it is left unchanged by the half-turn HV , then `id` and HV produce the same layout and the orbit contains only 2 distinct layouts rather than 4. This is precisely why we cannot

compute $N(n)$ by simply dividing the raw count $R(n)$ by 4: symmetric configurations sit in smaller orbits and would be undercounted.

The data in Table 7.1 confirms this. For example, $R(4) = 52$, yet $N(4) = 17 \neq 52/4 = 13$; the discrepancy is caused exactly by the layouts that are their own mirror images. Canonical labeling sidesteps this subtlety entirely, because it counts each orbit exactly once regardless of the orbit's size. In classical enumeration under group actions, the same count is obtained by Burnside's lemma, which computes the number of orbits as the average number of configurations fixed by each group element [7], [11]; our canonical-label method reaches the same number by direct, exhaustive enumeration instead.

7.2 Trends and Special Cases

There is a special case to pay attention to: **Square Grid $n = 3$** . As previously explained in Section 6, squares are special because they have double the symmetry of a typical rectangle (8 symmetry operations instead of 4). The computer finds a raw count of $R(3) = 2$ layouts. We can conclude that one layout is just a 90-degree rotation of the other. Since they belong to the same symmetry family, they share the same ID tag, resulting in $N(3) = 1$ unique configuration.

n	$R(n)$	$N(n)$	n	$R(n)$	$N(n)$
3	2	1	12	200,100	50,835
4	52	17	13	339,772	86,080
5	348	98	14	554,582	140,252
6	1,506	408	15	869,708	219,574
7	4,688	1,234	16	1,324,736	334,068
8	12,400	3,226	17	1,959,008	493,468
9	28,160	7,250	18	2,832,840	713,016
10	58,500	14,980	19	4,004,250	1,007,081
11	111,310	28,361	20	5,560,500	1,397,685

Table 1: Raw count $R(n)$ and symmetry-reduced count $N(n)$ of extremal configurations of $\Gamma_{3,n}$, for $3 \leq n \leq 20$.

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